



Influences of basic numerical abilities on graph reading performance

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Abstract

Understanding graphically presented information is an important aspect of modern mathematical and science literacy. In our study, we investigated the influence of basic numerical abilities on students' ability answer mathematical tasks with information presented in graphs. We analyzed data of 750 German students (grades 9–11) and evaluated the determinants of graph reading performance with multiple regression analysis using predictors of basic numerical abilities (such as number line estimation, basic arithmetic operations, etc.), considering also the influences of general cognitive ability, age, and gender. We found that number line estimation, subtraction, and conceptual knowledge were significant predictors of graph reading performance beyond the influences of general cognitive ability. This indicates that basic numerical abilities are still relevant for real-life problem solving in secondary school. We discuss possible mechanisms which directly (through respective arithmetic procedures) as well as indirectly (through mathematization of the problem) effectuate that basic numerical abilities graph reading performance.

Introduction

“Graphs, charts, cartograms, thematic maps, etc., are common tools for handling and communicating quantitative information in contemporary society.” (Åberg-Bengtsson, 2006, p. 112). During the last 10 years, quantitative data have played an increasingly important role in our life. It has become a daily routine to interpret the line diagrams displaying our heart rate data when doing sports, to read bar graphs in newspapers and infographics on TV. It is thus uncontroversial that reading quantitative graphs is an important twenty-first century skill (Ananiadou & Claro, 2009). Accordingly, many large-scale assessments of scholastic achievement refer to tasks involving quantitative graphs, for instance, within scientific literacy and mathematics tests

in TIMSS (Baumert, Bos, & Watermann, 1998) or reading, mathematical and science literacy assessments in PISA (Deutsches & Baumert, 2013). Strikingly, 60% of international 8th graders were only able to read a single value off a line graph and just 29% were able to determine the average of a graph (TIMSS, 2013). Due to the ubiquity and relevance of graphs in everyday life and education, it is worth investigating the factors that influence students' ability to fluently extract and use information from graphs.

The majority of research on graph reading addresses the influence of task (e.g., Lachmayer, 2008; Shah & Freedman, 2011) and graph (e.g., Wainer, 1992) characteristics (for reviews Friel, Curcio, & Bright, 2001; Shah & Freedman, 2011) on performance. In terms of the influence of graph reader characteristics, mathematical knowledge, experience with working on graphs, and general cognitive abilities have been identified as relevant factors for graph reading. However, to the best of our knowledge, there is no research evaluating graph reading abilities from the numerical cognition perspective.

Importantly, investigating influences of basic numerical abilities on graph reading allows for interconnecting two fields of research: (1) graph reading research that focuses on the ability to perform authentic everyday tasks with graphs, and (2) numerical cognition research which focuses on underlying processes, structure and the development of basic numerical abilities (e.g., understanding number

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magnitude, basic arithmetical operations, etc.). Interestingly, graph reading tasks involve working with visually, spatially and symbolically coded quantities. As such, measures of basic numerical abilities may allow to further differentiate individuals' abilities to work with these different codes of quantity.

Therefore, by investigating the influence of basic numerical abilities of graph reading, we may learn more about the underlying cognitive mechanisms that contribute to graph reading. In particular, the current study addresses the question of how basic numerical abilities influence graph reading performance. To the best of our knowledge, there is currently no other study investigating this association of basic numerical abilities and graph reading as a more applied ability. To sufficiently motivate this innovative new approach on the cognitive abilities underlying graph reading, we will first give a brief review of the literature investigating students' graph reading abilities before introducing the idea of basic numerical abilities serving as the building blocks of more advanced numerical and mathematical skills.

Graph reading

We define graph reading as the ability to fluently extract and use information from graphs. Graph reading is an important aspect for understanding information presented as graphs in everyday life (Galesic, & Garcia-Retamero, 2011). Various other terms have been used to denote similar constructs, such as graph literacy (Galesic, & Garcia-Retamero, 2011), graphicacy (Lowrie & Diezmann, 2011; Åberg-Bengtsson, 2006), graphing ability (Berg & Smith, 1994) and graph sense (Friel et al., 2001).

Generally, graph reading involves decoding and interpretation of visually displayed information. For Lowrie and Diezmann (2011), decoding and interpretation of bar and line graphs require knowledge about the graphical language. For bar and line graphs, it is the apposed-position language which encodes information by a mark positioned along both the x - and y -axes. To correctly interpret the coded information, it is necessary to integrate information from both axes. More specifically, Gillan and Lewis (1994) found that understanding a graph involves separate processing steps with non-arithmetic and arithmetic components. According to their Mixed Arithmetic-Perceptual Model, people interact with graphs to answer a certain question. They use search processes for indicators, encode the values of indicators, perform arithmetic operations based on these values, make spatial comparisons between the indicators, and finally respond to the question. The sequence of processing components may be different for each task. In the following, we argue that there are non-arithmetic and arithmetic processing components that seem to influence graph reading performance (cf. Gillan & Lewis, 1994).

Guthrie, Weber, and Kimmerly (1993) examined undergraduate students' understanding of graphs, tables, and illustrations. They found that two factors influenced their performance: (1) a—what they called—elementary level questions factor reflecting students' ability to locate specific information in the respective graphs and tables, and (2) an overall level questions factor referring to students' ability to perceive trends and patterns. The authors argued that perceiving trends and patterns requires abstraction processes which should be independent from the processes of locating specific information in graphs (Guthrie et al., 1993). Later on, Åberg-Bengtsson (1999) analyzed the underlying dimensions of performance on the diagrams, tables, and maps items of the Swedish Scholastic Aptitude Test. Besides a general factor that influenced performance on all items, she found a quantitative factor as well as a complexity factor. The general factor reflected aspects of locating the necessary information in the respective graphs or tables (e.g., along the x - and the y -axis of a line diagram, similar to the factor found by Guthrie et al., 1993) and thus non-arithmetic processing components that primarily include encoding. In contrast, the quantitative and the complex factor were more specific. The quantitative factor comprised all items that involved at least some calculations and thus arithmetic processing components. The third so-called complexity factor was associated with items that required multiple operations (numerical or not). The latter two factors may have reflected students' ability to "mathematize" the graph reading. With mathematizing, we refer to the students' ability to translate an ill-defined problem involving multiple steps into a mathematical structure (cf. Schoenfeld, 1989).

Additionally, there is evidence indicating that graph reading is significantly associated with general cognitive abilities. For instance, Berg and Smith (1994) found graph reading to be related to logical thinking and proportional reasoning in a sample of 7th, 9th, and 11th graders. Moreover, Padilla, McKenzie, and Shaw (1986) observed that interpreting line graphs was associated with abstract-reasoning abilities in a sample of 119 7th, 9th, and 11th graders.

Furthermore, there is also evidence suggesting that there may be gender differences in graph reading; boys were found to outperform girls (e.g., Lowrie & Diezmann, 2011; Åberg-Bengtsson, 1999; but see Curcio, 1987). Therefore, it is important to control for general cognitive abilities as well as gender when investigating the specific influence of basic numerical abilities on graph reading.

In sum, we suggest that students are facing different challenges when solving graph reading tasks. First, the respective graph reading problem needs to be mathematized into a corresponding mental model to enable them to—second—locate the relevant information, either numerical information or patterns and trends. The third challenge then is to

correctly perform the necessary arithmetic operations to answer the question at hand.

In line with this idea, Curcio (1987) found that 7th graders' comprehension of graphs and tables was associated positively with their prior knowledge about graphical language as well as about the mathematical content necessary to solve the tasks. This is first evidence suggesting that (basic) numerical abilities may specifically contribute to graph reading.

However, there are hitherto no empirical studies evaluating which specific basic numerical abilities influence graph reading performance. The current study pursued this question. Before describing the details of the present study, we will give a brief introduction to the idea of basic numerical abilities underlying numeric cognition.

Basic numerical abilities

Research has revealed that numerical cognition is not a unitary construct (e.g., Dowker, 2005). This means that numerical and mathematical skills build upon several basic numerical representations (e.g., Dehaene, 2009) and abilities, which are assumed to be the building blocks of numerical cognition in general and children's numerical development in particular (von Aster & Shalev, 2007). Basic numerical abilities include—but are not limited to—an understanding of non-symbolic and symbolic number magnitudes (e.g., Dehaene, Piazza, Pinel, & Cohen, 2003; Siegler, 2016) and a spatial representation of the corresponding magnitude (aka the mental number line, e.g., Booth & Siegler, 2008; see Fischer & Shaki, 2014 for a review), a verbal representation of number words, but also arithmetical facts (such as multiplication tables, e.g., Dehaene et al., 2003), a visual Arabic representation for understanding number symbols (e.g., Dehaene & Cohen, 1997; De Smedt, Noël, Gilmore, & Ansari, 2013 for a review), an understanding of the place-value structure of the Arabic number system (e.g., Moeller, Pixner, Zuber, Kaufmann, & Nuerk 2011; see Nuerk, Moeller, Klein, Willmes, & Fischer, 2011), as well as abilities on procedural and conceptual numerical knowledge (e.g., carry operation, but also commutative law).

A high level of mastery of these basic numerical abilities was repeatedly observed to be associated with actual numerical competencies but also predictive of future numerical competences as well as mathematical achievement in school (e.g., Moeller et al., 2011; Schneider, Grabner, & Paetsch, 2009).

In the following, a few examples will be given to illustrate how basic numerical abilities were found to influence later mathematical achievement. For instance, Kolkman, Kroesbergen, and Leseman (2013) observed that, already in kindergarten, children's early understanding of symbolic numbers as well as their skills in mapping these symbolic

numbers to non-symbolic magnitudes, such as dot patterns, was an important predictor of their later numerical/mathematical development (see Schneider et al., 2017 for a meta-analysis). Furthermore, Booth and Siegler (2008) found that children's performance on locating the position of a target number on an empty number line not only correlated positively with children's actual basic numerical abilities (e.g., magnitude comparison, see also Link, Nuerk, & Moeller, 2014), but also predicted their ability to acquire new arithmetical competences in the future. Related to this, Moeller et al. (2011) found that early place-value understanding in first grade predicted later arithmetic performance as well as children's math grades 2 years later. Moreover, recent evidence suggests that basic numerical abilities are predictive of understanding more complex mathematical concepts taught in secondary school such as fractions. For instance, Bailey, Siegler, and Geary (2014) found that basic arithmetic abilities assessed in first grade predicted arithmetic performance on fractions in secondary school (see also Vukovic et al., 2014). Interestingly, however, to the best of our knowledge, there is currently no study investigating the association of basic numerical abilities and graph reading performance. This comes somewhat as a surprise given the above-mentioned importance of graph reading in everyday life as well as in math curricula.

Therefore, the current study set out to evaluate the association of different basic numerical abilities with graph reading performance in a cross-sectional design. In particular, we aimed at assessing a battery of basic numerical abilities, which were proven to be relevant for numerical development, and to evaluate their prediction of concurrent graph reading performance in secondary school students. We hypothesized significant associations of such basic numerical abilities with graph reading which reflect (1) symbolic, visual, and spatial representations of magnitude (e.g., number line estimation, geometry) and (2) as well as arithmetic abilities necessary to carry out the necessary operations (e.g., addition, subtraction, but also conceptual knowledge). We expected such a pattern of results because graph reading tasks usually require a combination estimating/locating/encoding of number magnitudes and basic arithmetic operations performed on these magnitudes.

Method

Sample

The total sample included $N=812$ students from grades 9 to 11 recruited from all German school tracks (including academic as well as vocational track schools and vocational schools). We choose this sample as it provided the most representative cohort of students of that age, from those in

vocational schools to those preparing for their Abitur (i.e., A-levels) in academic track schools. After outlier inspection for age, we excluded 22 students older than 25 years of age from the analysis. These were from vocational schools. The vast majority of students enter vocational schools after completing grades 9 or 10 at other school tracks. However, some students start vocational education later (e.g., after parental leave). Furthermore, we excluded 15 students, because of missing demographic data, and 25 students because they did not complete the graph reading test. This left 750 students for the analyses. The final sample had a mean age of $M = 17.34$ years ($SD = 2.12$) and included 36% female students.

Measures

Graph reading test

Graph reading was assessed by 14 items, seven of which addressed a bar and seven a line graph. Respective responses were either a specific number or a specific label of the graph. The time for doing the whole test was restricted to 4 min. Correctly answered items were considered in a sum score, which served as the dependent variable.

Successful completion of the graph reading items required locating specific information in the graph, several numerical operations such as magnitude comparisons, basic arithmetical operations, as well as more complex modeling demands. An item was subject to modeling demands when it required an inference about which arithmetic operation had to be performed. For instance, to correctly answer a

modeling item, participants had to infer that they would have to calculate the sum of two values.

Figure 1 gives an example of a bar graph. An exemplary item might have been “Which worker should be rewarded?”. Students first needed to consider the context of the problem to infer that the worker who produced the highest quantity of shoes should be rewarded, (modeling demands), second, determine the worker with the longest bar graph (relational operation), third, sum up the two bars of potential candidates (addition) by reading off the exact values (localization of information), and fourth, read the label to identify the worker who had produced the most shoes (localization of information). Please note that the item given in Fig. 1 is for illustrative purposes only and was not an item actually used in the test.

Basic numerical abilities

Different basic numerical abilities—ranging from non-symbolic magnitude comparisons to (approximate) arithmetic—were assessed by eight separate tests. We aimed at assessing different basic numerical abilities for which previous research indicated their relevance for numerical development and mathematics achievement. Selection of tests did not follow a specific model of numerical development and we do not wish to claim that the list of basic numerical abilities assessed is comprehensive or exhaustive.

All eight tests were comprised of materials which primary schools students should be able to solve. Therefore, the number of solved items within the given time allows conclusions on the degree of automatization. For all tests, students were presented with examples to familiarize themselves with the topics. Unless indicated differently, correctly solved items were considered for sum scores which were used as the dependent variable in all analyses. In the following, we will describe the tests used in more detail.

Non-symbolic magnitude comparisons Students were each presented with 16 pairs of dot clouds and had to mark the cloud with the larger quantity of dots. The overall surface of the dot clouds was systematically varied in a way that for half of the items the surface with the lower quantity of dots was larger, while for the other half of the items, this was reversed. The time for this task was limited to avoid counting-based strategies (time limit: 1 min). Non-symbolic number magnitude comparison is the primary task to assess the so-called Approximate Number System, which is considered an innate basis for later numerical and mathematical development (e.g., Feigenson, Dehaene, & Spelke, 2004; Halberda, Mazzocco, & Feigenson, 2008; Park & Brannon, 2013).

Number line estimation In the number line estimation task, students had to estimate the spatial location of a given number on a number line, of which only the endpoints were

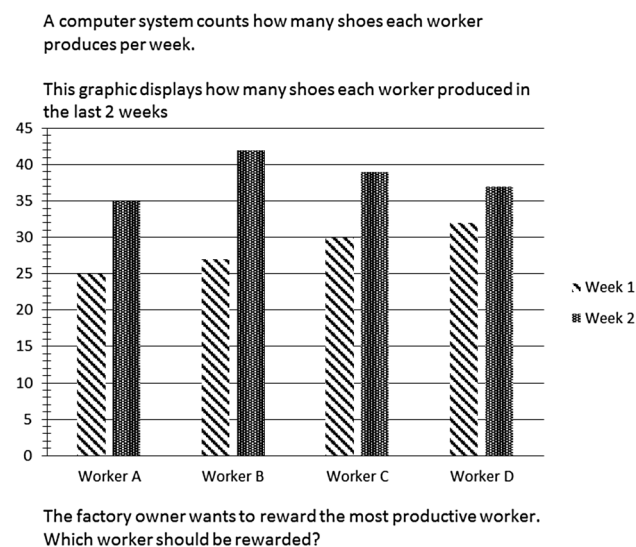


Fig. 1 Schematic illustration of item format introducing item stem with a sample item in open answer format. Please note that this is not an item used in the experiment

specified (e.g., the spatial location of 24 on a number line ranging from 0 to 100). Overall, the task comprised 24 items (six items on a number line from 0 to 10 and 0 to 100, respectively, and four items each on a number line from 0 to 1.000, 0 to 10.000, and 0 to 100.000, respectively). The time for the task was limited to 1.5 min to ensure that students estimated the position of the target numbers intuitively. The mean absolute estimation accuracy in percent served as the dependent variable and was used in the analyses. Items with no response were coded with an accuracy of zero. Number line estimation is considered to assess a spatial representation of number magnitude (e.g., Siegler & Opfer, 2003, but see Barth & Paladino, 2011) and was repeatedly observed to predict actual but also future numerical and mathematical performance (e.g., Booth & Siegler, 2008; Link et al., 2014).

Arithmetic operations Arithmetic operations included: (1) addition, (2) subtraction, and (3) multiplication. Each test was comprised of 36 arithmetic tasks in the order of ascending difficulty. Multiplications were restricted to single digit times single digit tasks, whereas addition and subtraction tasks covered the number range up to 10,000. Out of each operation, students had to solve as many tasks as they could within 2 min. Basic arithmetic operations are at the core of mathematical development and achievement (e.g., Ashcraft, 1982; Geary, 2011; Price, Mazzocco, & Ansari, 2013).

Approximate arithmetic In the approximate arithmetic tasks, students had to choose the one out of two incorrect solution proposals that were closer to the correct result. The range comprised 16 addition and 16 subtraction tasks. For instance: “Which result is closer to 1546–687? Possible answers: 816 or 678”. As in the basic arithmetic tests, the difficulty level increased with every item. Students had 2 min to solve the tasks. Approximate arithmetic is repeatedly observed to underlie and predict mathematics achievement (e.g., Gilmore, McCarthy, & Spelke, 2007; Gilmore, McCarthy, & Spelke, 2010).

Basic geometry Basic geometrical abilities were measured using 12 mirror image problems. Students were presented with a geometric shape (e.g., a rectangle) and an axis across which they had to mirror this shape by drawing the flipped form on the side opposite the mirror axis. For each correct line in a drawing, students were assigned one point. As a result, maximum scores for each item varied between 6 and 12. This resulted in a maximum score of 94 for this scale. A sum score was used as the dependent variable. Basic geometry is an important area of mathematics learning (e.g., Clements & Sarama, 2011; Clements, Wilson, & Sarama, 2004), with the present symmetry task being used in standardized tests for mathematics achievement (e.g., Götz, Lingel, & Schneider, 2013).

Conceptual knowledge about arithmetic In this task, students were presented with 40 pairs of arithmetic problems containing addition, subtraction, multiplication, and

division. In each pair, one of the two tasks was already solved. Students had to decide whether the solution to the first problem helps them to solve the second problem without having to calculate. For example: (a) “Does 54: 9 = 6 help you solve 54: 6 = ?” or (b) “Does $4 \times 23 = 92$ help you solve $92: 4 = ?$ ”. Conceptual knowledge represents “knowledge of general numerical principles and knowledge of principles underlying procedures” (p. 344, Crooks & Alibali, 2014; see also Rittle-Johnson, Schneider, & Star, 2015), which are assessed using this task.

Additionally, we assessed children’s general cognitive ability as a covariate with two subtests of the German version of the Culture Fair Intelligence Scale 2-revision (i.e., sequence continuation, matrices; CFT 20-R; Weiß, Albinus, & Arzt, 2006). Subtests were administered as defined in the manual. In the sequence continuation subtest, students needed to find a logical continuation of a sequence of shapes. In the matrix subtest, students needed to make the same kind of conclusion for finding a logical shape for a blank cell of a matrix.

Procedure

All tests were administered during school hours in the students’ classrooms by trained experimenters. Testing took maximally 90 min. Parents provided informed written consent, and students were told that they could withdraw from the test at any time without negative consequences. Students above the age of 18 provided informed written consent by themselves, and parents received information about the study. The study was approved by the local ethical committee as well as the regional school board.

Statistical analyses

Prior to running the analyses, we checked for multicollinearity between variables. However, multicollinearity was no issue, because no two variables correlated higher than $r_{ij} > 0.8$ and no predictor variable showed a variance inflation factor (VIF) > 10 (cf. O’Brien, 2007). Moreover, all variables except age were approximately normally distributed. The distribution of age was skewed, due to the few students who were older than is usually expected for students in the investigated school types. Therefore, we log transformed the age variable prior to the analyses.

Multiple regression

We used multiple regression analysis to determine the significant predictors of graph reading performance with a false discovery rate (FDR) p value adjustment for multiple testing (Benjamini & Hochberg, 1995).

Relative weight analysis

Additionally, we reported the relative weight of each predictor. The relative weight analysis (Johnson, 2000) addresses a problem caused by correlated predictors. Relative weight analysis uses a variable transformation approach to create a set of new predictors that are maximally related to the original predictors but are orthogonal to one another. In contrast to standardized regression weights, resulting relative weights represent the predictors' additive decomposition of the total model R^2 . Two measures of relative weight can be calculated, the raw relative weight and the rescaled relative weight. Raw relative weights add up to the R^2 of the model and the rescaled relative weights add up to 100%, representing the relative importance of a particular variable in the regression model. Relative weights can be interpreted as the proportion of explained variance in criteria that can be appropriately attributed to each predictor variable (Tonidandel & LeBreton, 2015). Finally, we determined the significance of relative weights using the procedure described by Tonidandel, LeBreton, and Johnson (2009).

Variables

For all analyses, we considered 29 predictor variables: the eight basic numerical abilities that were assessed (i.e., addition, subtraction, multiplication, number line estimation, approximate arithmetic, conceptual knowledge, basic geometry, non-symbolic magnitude comparisons), as well as general cognitive ability, age, gender, and the interaction terms of age and gender with general cognitive ability and the eight basic numerical abilities.

We used an effect coding for gender (-1 = female, male = 1) and centered all continuous variables to be able to interpret the effect of interaction terms.

Statistical software

All statistical analyses were performed in the R environment (R Core Team, 2017). Multiple regression analysis was performed using the 'lm' function for fitting the linear models of the standard R package "state" (R Core Team, 2017). We used the 'p.adjust' function with the 'fdr' method to adjust p values. Finally, we applied the syntax adopted from Tonidandel and LeBreton (2015) to conduct the relative weights analysis.

Results

The average graph reading score was $M = 6.71$ ($SD = 2.28$) out of 14 possible points. Therefore, graph reading showed sufficient variability. The predictor variables also showed

Table 1 Mean (M), standard deviation (SD) and obtained range of all measures

	M	SD	Range
Graph reading	6.75	2.28	1–12
Addition	20.86	4.23	2–31
Subtraction	17.17	4.96	0–32
Multiplication	20.94	4.32	2–29
Number line estimation (%)	81.27	15.77	2.19–97.47
Approximate arithmetic	20.40	5.33	0–32
Conceptual knowledge	18.21	6.10	1–36
Basic geometry (%)	59.35	21.18	0.00–100.00
Non-sym. mag. comp.	18.75	3.13	1–24
G. cognitive ability	19.01	4.11	1–29

$N = 750$

sufficient variability (see Table 1), indicating that under the given time constraints none of the measures were either too easy or too difficult.

The correlation matrix depicted in Table 2 indicated that almost all basic numerical abilities were significantly correlated with graph reading performance, as well as among each other. This held with only one exception: conceptual knowledge and non-symbolic magnitude comparisons were not related significantly. The control variables gender and age showed small correlations with some measures of basic numerical abilities (see Table 2). Gender was significantly correlated with graph reading performance. Age was negatively related to conceptual knowledge, basic geometry, and general cognitive ability. These significant correlations justified our decision to consider gender and age as control variables when evaluating the influence of basic numerical abilities on graph reading performance.

Multiple regression analysis The linear multiple regression analysis including 29 predictors (i.e., basic numerical abilities, general cognitive ability, age, gender, and the interactions of basic numerical abilities with age and gender) which explained about 34% of the variance [$R^2 = 0.34$, adj. $R^2 = 0.31$, $F(30, 720) = 12.61$, $p < .001$] of graph reading performance. Four individual variables showed a significant effect on graph reading performance: general cognitive ability, number line estimation, subtraction, and conceptual knowledge (see Table 3). Inspection of the beta weights indicated that for all predictors better performance on the predictor was associated with better graph reading performance. General cognitive ability had a considerably larger effect on graph reading performance than conceptual knowledge because the 95% confidence intervals for the respective coefficients did not overlap. However, effect sizes of all other predictors were indistinguishable.

Additionally, we computed the relative weight of each variable (Johnson, 2000) to evaluate which predictors

Table 2 Correlations between graph reading, basic numerical abilities, and general cognitive ability, as well as gender and age

Variable	1	2	3	4	5	6	7	8	9	10	11
1. Graph reading	1										
2. Addition	0.31**	1									
3. Subtraction	0.39**	0.68**	1								
4. Multiplication	0.31**	0.58**	0.58**	1							
5. Number line estimation	0.29**	0.25**	0.22**	0.21**	1						
6. Approximate arithmetic	0.29**	0.40**	0.44**	0.36**	0.37**	1					
7. Conceptual knowledge	0.31**	0.32**	0.33**	0.31**	0.34**	0.39**	1				
8. Basic geometry	0.24**	0.23**	0.21**	0.15**	0.09**	0.08*	0.20**	1			
9. Non-sym. mag. comp.	0.14**	0.15**	0.10*	0.14**	0.06*	0.12**	0.07	0.08*	1		
10. G. cognitive ability	0.45**	0.34**	0.35**	0.32**	0.17**	0.21**	0.28**	0.41**	0.24**	1	
11. Gender ^a	0.10**	0.11**	0.23**	0.14**	0.24**	0.21**	− 0.08*	− 0.06	− 0.06	0.00	1
12. Log (age)	− 0.05	0.06	0.07	− 0.05	− 0.03	0.05	− 0.12**	− 0.10**	0.03	− 0.17**	0.25**

** $p < .01$, * $p < .05$. $N = 750$

^aCode female = − 1, male = 1

Table 3 Multiple regression results

	<i>B</i>	β	[<i>L-CI</i> , <i>U-CI</i>]	<i>RW</i>	<i>t</i>	<i>p</i>	<i>RS-RW</i> (%)
Criteria = Graph reading performance [multiple $R^2 = 0.34$, <i>adj. R</i> ² = 0.31, $F(30, 720) = 12.61$, $p < 0.001$]							
Intercept	6.71	0.00	[− 0.06, 0.07]		80.53	0.000	
G. cognitive ability	0.17	0.30	[0.22, 0.37]	0.10	7.34	0.000	28.54*
Subtraction	0.09	0.20	[0.11, 0.29]	0.05	4.17	0.000	13.86*
Number line estimation	2.13	0.15	[0.06, 0.19]	0.04	4.07	0.000	12.01*
Conceptual knowledge	0.04	0.11	[0.03, 0.18]	0.03	2.86	0.026	9.92*
Approximate arithmetic	0.01	0.03	[− 0.04, 0.11]	0.02	0.80	0.748	6.12*
Multiplication	0.01	0.01	[− 0.05, 0.11]	0.02	0.34	0.914	6.22*
Basic geometry	0.01	0.05	[− 0.03, 0.11]	0.02	1.27	0.616	5.17*
Addition	− 0.01	− 0.02	[− 0.14, 0.04]	0.02	− 0.35	0.914	5.79*
Non-sym. mag. comp	0.01	0.01	[− 0.04, 0.09]	0.01	0.29	0.925	1.70
Gender ^a	0.07	0.03	[− 0.04, 0.10]	0.00	0.87	0.748	1.16
Log (age)	0.14	0.01	[− 0.06, 0.07]	0.00	0.22	0.954	0.46

Please note that for reasons of better readability only main effects are displayed (see Appendix for a table including interaction terms)

b unstandardized regression weight, β standardized regression weight, *L-CI* lower boundary (2.5%), *L-CI* upper boundary (97.5%), *RW* raw relative weight (within rounding error raw weights will sum to R^2), *t* *t* value measures the size of the effect relative to the variation in sample data, *RS-RW* relative weight rescaled as a, percentage of predicted variance in the criterion variable attributed to each predictor (within rounding error rescaled weights sum to 100%)

*Significantly different from a random variable

^aCode female = − 1, male = 1

accounted for non-trivial variance of graph reading performance in contrast to regression weights that reflect incremental prediction. In case predictors are correlated they may not yield a significant incremental relationship.

Rescaled relative weights revealed that general cognitive ability accounted for a share of 28.54% of the explained variance in graph reading performance. Therefore, general cognitive ability seemed to be the best predictor for graph reading performance. Interestingly, the sum of basic numerical

abilities accounted for a larger share of the explained variance in graph reading performance.

Furthermore, despite low regression weights, consideration of rescaled relative weights indicated that addition ($\beta > -0.02$), multiplication ($\beta = 0.01$), approximate arithmetic ($\beta = 0.03$), and basic geometry ($\beta = 0.05$) explained a significant proportion of the predicted variance (i.e., between 5 and 6%). This showed that these skills may be relevant to the prediction of graph reading performance, but there was

not enough incremental variance in the regression model to become significant. In contrast, gender and age had similarly low standardized regression weights (gender: $\beta=0.01$; age: $\beta=0.01$), but also very low rescaled relative weights (gender: RS-RW = 1.01; age: RS-RW = 0.42). Therefore, they may not be as relevant for the prediction of graph reading performance.

Discussion

In the current study, we aimed at investigating the influence of basic numerical abilities on graph reading performance. In two separate analyses, we first identified the significant predictors of graph reading performance and then evaluated the relative importance of individual basic numerical abilities. We observed general cognitive ability to be the most important predictor of graph reading performance. Beyond general cognitive ability, performance in number line estimation, subtraction, and conceptual knowledge were significant predictors of graph reading performance. This finding is in line with the Mixed Arithmetic-Perceptual Model by Gillan and colleagues (Gillan & Lewis, 1994; Gillan, 2009) as the significant predictors are associated with arithmetical processes (subtraction and conceptual knowledge), and non-arithmetical processes (number line estimation and general cognitive ability). Furthermore, they also map well with Åberg-Bengtsson (1999) differentiation between a general factor (reflected by general cognitive ability) and a quantitative factor (involving subtraction, number line estimation, and conceptual knowledge).

In the following, we will discuss these results in more detail by addressing each of the relevant predictors of graph reading performance one after the other. In particular, we will consider two questions: How may the respective predictor be related to graph reading (e.g., identifying specific information, relational operations, etc.) and which general cognitive processes and cognitive strategies may underlie its influence?

The most important predictor of graph reading was general cognitive ability. This influence of general cognitive ability is not surprising; however, there may be two reasons why general cognitive ability may have been of particularly predictive value for graph reading performance. First, graph reading problems may not be a common part of the daily routine in schools. General cognitive ability is often defined as the ability to solve new problems (e.g., Hartig, & Klieme, 2006). Therefore, general cognitive ability was found to be a significant predictor of graph reading performance. Second, the subtests of the Culture Fair Test used in the present study strongly rely on visual processing and analogical reasoning (Weiß et al., 2006). These processing components also relate

to the non-arithmetical processes, and the visual imagery processes described in the Mixed Arithmetic-Perceptual Model by Gillan and Lewis (1994). Importantly, similar subcomponents of information processing have also been discussed to influence graph reading by other authors (cf. Verschaffel, De Corte, & Lasure, 1994).

Moreover, number line estimation was observed to be the most important basic numerical ability to predict graph reading performance: Better number line estimation performance predicted better graph reading. Interestingly, number line estimation involves visuo-spatial processes, such as performing spatial comparisons, value encoding, and spatial differences. These components are described in the Mixed Arithmetic-Perceptual Model (Gillan & Lewis, 1994; Gillan, 2009). This demonstrates that components, which have been identified by Gillan and Lewis (1994) based on response time experiments and think loud protocols, match components reflected by predictors of graph reading of the current study.

Furthermore, number line estimation is the only basic numerical ability—among those predictors considered significant for graph reading—that involves translations between symbolically and spatially coded representations of quantity. This again is specifically required in graph reading. Moreover, localizing specific information in a spatial graphical set-up is a core component of graph reading (Guthrie et al., 1993) which usually necessitates processing of quantitative information across different representational modalities. For instance, reading a specific value off a graph requires identifying the location of the respective information (e.g., the largest bar), and then to translate this spatial representation of magnitude into a symbolic representation by reading the corresponding position off a scale.

Additionally, recent research (e.g., Barth & Paladino, 2011; see Dackermann, Huber, Bahnmueller, Nuerk, & Moeller, 2015 for a discussion) has shown that number line estimation usually involves the application of so-called proportion-judgment strategies. This means that participants estimate the position of a target number by considering its relation to specific reference points (e.g., the middle of the depicted number line, i.e., 50 on a 0–100 number line). Very similar proportion-based strategies were observed to be applied for the estimation of bar graph lengths (Hollands & Spence, 1998). Furthermore, most of the present graph reading tasks involved relational operations (i.e., using equal to, greater than, smaller than). Accordingly, students who performed well in number line estimation due to their ability to translate between symbolic and non-symbolic spatial representations of quantity and/or the application of efficient proportion-based strategies might also have performed better in graph reading tasks.

As another numerical predictor, subtraction was identified as a relevant and significant predictor of graph reading

performance with better subtraction performance being associated with better graph reading. Most graph reading tasks in the present study involved arithmetic operations. However, subtraction was the only relevant and significant predictor of graph reading. There may be at least three reasons for that. First, variability was more pronounced for subtraction as compared to addition, which might account for the fact that subtraction and not addition was incorporated in the variable selection regression model. Second, addition is more prone to influences of arithmetic fact retrieval, which has been suggested for single digit additions (e.g., Butterworth, Zorzi, Girelli, & Jonckheere, 2001; Pesenti, Thioux, Seron, & De Volder, 2000). In turn, the subtraction scale may better reflect students' procedural calculation skills. This argument is further corroborated by the fact that multiplication performance, which is widely agreed to reflect arithmetic fact retrieval (e.g., Butterworth et al., 2001; Pesenti et al., 2000), was not observed to be a significant predictor of graph reading. Finally, subtraction and addition performance were highly correlated. Considering the first two arguments, it is unlikely that addition explains unique variance over and above what had already been explained by subtraction.

Finally, students' conceptual knowledge was a significant predictor of their graph reading in a way that better performance in the conceptual knowledge scale was associated with better graph reading performance. In the conceptual knowledge scale, students needed to identify relationships between arithmetic operations rather than solve arithmetic problems by actually performing the necessary computations. As such, conceptual knowledge enabled students to select and use problem solving strategies which are less resource demanding. For instance, in some tasks, students had to compute the means of two values depicted in a bar graph. The mean can be determined by reading the values off the two bars and then applying the arithmetic procedure $[M = (x_1 + x_2 + \dots + x_n)/n]$. However, knowledge about relationships between arithmetic operations enables students to select a problem solving strategy demanding fewer resources. Such a strategy could be to take the spatial middle between the two respective graphs and then read off the respective value (Gillan, 1995). Therefore, higher conceptual knowledge may enable students to apply more efficient problem solving strategies in graph reading problems.

As indicated by the relative weight analysis multiplication, approximate arithmetic, and basic geometry accounted for a considerable amount of predicted variance of graph reading performance and might thus be relevant predictors of it as well. However, their influence turned out not to be significant in the multiple regression analysis—most probably because they share large parts of their variance with other predictors included in the model.

Multiplication may be a generally relevant predictor because of the influence of arithmetic fact retrieval (i.e.,

multiplication tables), which describe a different way of solving numerical tasks as compared to actual calculations. However, none of the problems in the graph reading test explicitly required multiplication. Moreover, approximate arithmetical abilities may be a relevant predictor because students with good approximation abilities may have advantages in judging the plausibility of possible results. However, all problems of the graph reading test required that students provided a specific numerical result, which could not be solved by approximation alone. Finally, basic geometry may be relevant for graph reading due to the necessity to process spatially represented information and may relate different pieces of spatial information explicitly to each other (e.g., to evaluate which worker produced more shoes by comparing the size of the bars, cf. Fig. 1). All that is reflected in the basic geometry task, which also required participants to process spatial information and relate various pieces to produce the respective mirror images. Therefore, it may be necessary that these basic numerical abilities be acknowledged even though they were not identified as significant predictors of graph reading performance within the set of basic numerical abilities assessed in the current study.

So, what were the basic numerical predictors of graph reading? We identified number line estimation and subtraction performance as well as students' conceptual knowledge to be significant predictors of graph reading beyond the influence of general cognitive ability. Interestingly, these basic numerical abilities seem to be associated with graph reading because they reflect specific numerical processes and more general problem solving strategies required in graph reading. In fact, the set of significant predictors identified in the present study also fit the differentiation of arithmetical and non-arithmetical cognitive components contributing to graph reading performance as suggested by the Mixes Arithmetic-Perceptual model of graph reading by Gillan and Lewis (1994). Yet, the influences of predictors involving computational skills such as subtraction seem obvious. Similarly, number line estimation implies processes such as the translation between symbolically and spatially coded representations of quantities that are also necessary for graph reading tasks.

However, considering how number line estimation is achieved (by means of proportion-based strategies) and the observed influence of students' conceptual knowledge, we conclude that the influence of basic numerical abilities goes beyond the level of providing mere numerical and arithmetical prerequisites for more applied everyday graph reading. Instead, the influence of number line estimation and conceptual knowledge indicates that the mastery of basic numerical concepts allows for the application of more efficient problem solving strategies beyond mere arithmetical procedures.

Conclusions and perspectives

We observed the influence of basic numerical abilities on graph reading over and above influences of general cognitive ability. To the best of our knowledge, this is one of the first cross-sectional studies indicating that students, who score high on basic numerical abilities acquired in primary school, also perform better on more applied everyday graph reading tasks at a secondary school level. Additionally, basic numerical abilities and general cognitive ability, which we identified as significant predictors of graph reading, closely map with the distinction between arithmetical and non-arithmetical processes described by the Mixed Arithmetic-Perceptual Model of graph reading (Gillan & Lewis, 1994).

In sum, the present results underline the central importance of basic numerical abilities that are developed before and during the first years of formal schooling and later their impact on more complex math skills, which, in turn, are relevant for educational achievement and everyday life. We are faced with different forms of graphs in various situations every day (Åberg-Bengtsson, 2006), for instance regarding the results of elections, stock market development, labor market information, and more. As indicated by the results of this study, the foundation for graph reading is laid in primary school. However, by secondary education, teachers usually expect students to have acquired a sufficient level of basic numerical abilities. Therefore, teachers do not focus on them again. As a consequence, students with deficits in these areas may experience severe problems in solving more complex tasks such as graph reading. Even though the present results rely on cross-sectional data, they indicate that even in secondary education, it seems worth reviewing elementary school knowledge in the context of real-world math problems.

Nevertheless, it would be desirable to replicate the observed association between basic numerical abilities and graph reading in longitudinal studies or even find beneficial effects of training basic numerical abilities on graph reading performance in intervention studies—the latter allowing for causal interpretations of the association. Furthermore, it is important to note that we only controlled for influences of general cognitive ability as influences of domain-general cognitive abilities. Of course, it would be interesting and necessary to evaluate the influences of other domain-general cognitive abilities such as working memory and/or visuospatial abilities to substantiate the observed contribution of specific numerical abilities to graph reading.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Human and animal rights statement All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Appendix

Results of multiple regression analysis and relative weights of basic numerical abilities, general cognitive ability, age, gender and two-way interaction between age, gender with basic numerical abilities and general cognitive ability.

	<i>B</i>	β	[L-CI,U-CI]	RW	<i>t</i>	<i>p</i>	RS-RW (%)
Criteria = Graph reading performance [multiple $R^2 = 0.33$, adj. $R^2 = 0.31$, $F(30,720) = 12.46$, $p < 0.001$]							
Intercept	6.71	0.00	[− 0.06, 0.07]		80.53	0.000	
Addition	− 0.01	− 0.02	[− 0.14, 0.04]	0.02	− 0.35	0.914	5.79*
Subtraction	0.09	0.20	[0.11, 0.29]	0.05	4.17	0.000	13.86*
Multiplication	0.01	0.01	[− 0.05, 0.11]	0.02	0.34	0.914	6.22*
Number line estimation	2.13	0.15	[0.06, 0.19]	0.04	4.07	0.000	12.01*
Approximate arithmetic	0.01	0.03	[− 0.04, 0.11]	0.02	0.80	0.748	6.12*
Conceptual knowledge	0.04	0.11	[0.03, 0.18]	0.03	2.86	0.026	9.92*
Basic geometry	0.01	0.05	[− 0.03, 0.11]	0.02	1.27	0.616	5.17*
Non-sym. mag. comp.	0.01	0.01	[− 0.04, 0.09]	0.01	0.29	0.925	1.70
G. cognitive ability	0.17	0.30	[0.22, 0.37]	0.10	7.34	0.000	28.54*
Gender ^a	0.07	0.03	[− 0.04, 0.10]	0.00	0.87	0.748	1.16
Log (age)	0.14	0.01	[− 0.06, 0.07]	0.00	0.22	0.954	0.46
Addition × age	0.2	0.04	[− 0.05, 0.14]	0.00	0.89	0.748	1.31
Subtraction × age	− 0.15	− 0.04	[− 0.13, 0.05]	0.00	− 0.82	0.748	0.41
Multiplication × age	− 0.01	0.00	[− 0.09, 0.08]	0.00	− 0.06	0.954	0.22
Number line estimation × age	2.56	0.02	[− 0.05, 0.09]	0.01	0.59	0.828	2.26

	<i>B</i>	β	[<i>L</i> - <i>CI</i> , <i>U</i> - <i>CI</i>]	<i>RW</i>	<i>t</i>	<i>p</i>	<i>RS</i> - <i>RW</i> (%)
Approximate arithmetic $\times$ age	− 0.07	− 0.02	[− 0.09, 0.05]	0.00	− 0.54	0.847	0.54
Conceptual knowledge $\times$ age	− 0.11	− 0.03	[− 0.11, 0.04]	0.00	− 0.92	0.748	0.34
Basic geometry $\times$ age	0.04	0.04	[− 0.03, 0.11]	0.00	1.12	0.713	0.31
Non-sym. mag. comp. $\times$ age	0.18	0.03	[− 0.04, 0.10]	0.00	0.83	0.748	0.61
G. cognitive ability $\times$ age	− 0.11	− 0.03	[− 0.09, 0.05]	0.00	− 0.65	0.828	0.80
Addition $\times$ gender	− 0.06	− 0.11	[− 0.2, − 0.01]	0.00	− 2.16	0.155	0.16
Subtraction $\times$ gender	0	0.00	[− 0.09, 0.10]	0.00	0.08	0.954	0.12
Multiplication $\times$ gender	0.03	0.06	[− 0.03, 0.13]	0.00	1.33	0.616	0.13
Number line estimation $\times$ gender	− 1.08	− 0.07	[− 0.14, 0.00]	0.00	− 2.04	0.178	0.09
Approximate arithmetic $\times$ gender	0	0.00	[− 0.08, 0.09]	0.00	0.07	0.954	0.14
Conceptual knowledge $\times$ gender	− 0.01	− 0.02	[− 0.09, 0.06]	0.00	− 0.46	0.877	0.57
Basic geometry $\times$ gender	0	− 0.02	[− 0.10, 0.05]	0.00	− 0.63	0.828	0.51
Non-sym. mag. comp. $\times$ gender	0.03	0.05	[− 0.02, 0.11]	0.00	1.29	0.616	0.44
G. cognitive ability $\times$ gender	0	− 0.01	[− 0.08, 0.07]	0.00	− 0.13	0.954	0.07
Σ	−	−	−	0.33	−	−	100.00

B unstandardized regression weight, β standardized regression weight, *L*-*CI* lower boundary of 95%-confidence interval, *U*-*CI* upper boundary of 95%-confidence interval, *RW* raw relative weight (within rounding error raw weights will sum to R^2), *t* *t* value measures the size of the effect relative to the variation in sample data, *p* FDR adjusted *p* value, *RS*-*RW* relative weight rescaled as a percentage of predicted variance in the criterion variable attributed to each predictor (within rounding error rescaled weights sum to 100%)

*Significantly different from a random variable

^aCode female = − 1, male = 1

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